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Clickboat: Boat repair and rental Management System

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RATIONALE

The nature of boat repair and rental counting are time consuming and requires workforce and more effort to do. Manual process may also be hard to be done on a regular basis. By using the proposed project, boat repair and rental transaction processes will be automated and will be easy to manage on a daily and regular basis. The yacht owner, yacht charter, tourist and boat owners will be able to save time and effort such as notifications and updating transaction status.

OBJECTIVES

The general objective of the study is to develop a web/mobile application that can be used by yacht owner, yacht charter, tourist and boat owners. To be able to understand the main objective, the specific objectives were drawn: (1) Define problems encountered in: a. Yacht repair scheduling ,b.Yacht repair management, c. tourist iterinary ,d. boat scheduling;(2) Define modules of the system developed to address the problem;(3) Define the tools and technology used to develop the solution; (4) the level of acceptance of the respondents in the developed system in terms of: a. user friendliness , functionality ; (5) Issues encountered during the initial deployment of the system;(6) Features recommended to improve the system.

METHODOLOGY

To be able to develop the final output, the study followed the processes involved in the Agile Development. It includes all the processes that is necessary to finish the study which are: (1)Plan , (2)Design, (3)Develop (4)Test, (5) Release , (6) Feedback. The data needed to conduct the study were analysed and validated based from the system requirements of the project.

RESULTS

The system's evaluation resulted with the mean range of 4 above with the equivalent of high acceptability rating for both user friendliness and functionality. The developed modules are for tourist, yacht charter, yacht owner and boat owner where they can communicate and automate some of the process that involves notification and report generation.

CONCLUSIONS

Mobile and web application is a good way to cater marina based services such as boat repair and rental. Different available API and libraries are very useful in the development because it speeds up the process. The ability to test different environment of the system is a must especially technology is evolving.

KEYWORDS:*Yacht charter, Tourist Itinerary, Web application, Boat Scheduling, Yacht repair scheduling*.

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1. INTRODUCTION

The boating and yacht industry has been one of the major components in terms of marina tourism, Boat rental and repair has been vital yacht charter maybe not a typical type of business even though owning yacht is pretty expensive clients are still availing and finding services for boat repairs, same as the boat rental industry which may comprises tourism such as boat ride and island hopping , in addition to that Philippines is an archipelago that makes it a good spot for marina services such as marina tourism where the Tourist can book boat ride on Boat owners for tourists spot islands such as Fortune Island and Twin Island on Nasugbu,Batangas. The Country is also a hub for numerous number of Yacht charter that offers repairs on either boat or ships. Even though in some countries like US there are unemployment rate increase that makes the market on boating to decline but still the industry revenue manages to come back on the market with some constraints on the line such as smaller market but few competitors around due to unemployment of the crews.

Papaya yacht charter is one of the local yacht charter that offers this kind of service on the municipality of Nasugbu, they often use paper and flat database like excel on recording transaction, because of this report generating can take time to do and another is the handling of transactions is hard when it is open season or occasion of yacht repairs and maintenance transactions record are hard to keep because it is on the form of paper records that is probable of it to lost and will not be able to retrieve it. Another is the Boat owners have difficulty in terms of managing the transactions in terms of different tourist, tourist may also encounter difficulties of looking boat owners who has the right size of boat to rent. Barangay Papaya has been a tourist spot on the past years since 2015, manual transaction has been the constraint they have been facing that makes it difficult for the boat owners to cater guests.

The purpose of this research titled “Click Boat : Boat Repair and Rental Management System” is to improve and the business process on the industry of Boat repair and rental with the use of technologies and develop a system to serve both sides of yacht owners,manager, boat owner and tourists to help improve the aspects of the business such as report generating, tracking and management of the information needed to became a functional and efficient service provider. .

2. OBJECTIVES

The research aim to develop ClickBoat: Boat Repair and Rental Management System.
Specifically it sought the following:

1. What are the common problems encountered by the respondents in terms of:
   - 1.1 Tourist’s Itinerary
   - 1.2 Boat Scheduling
   - 1.3 Yacht Repair Scheduling
   - 1.4 Yacht Repair Management
2. What are the modules of the system to be develop to address the common problems encountered by the respondents?
3. What are the tools and technology did the Researchers use to develop the solution to the problem?
4. What is the level of acceptance of the respondents in the developed system “Click Boat: Boat Repair and Rental Management System” in terms of?
   - 4.1 User Friendliness; and
   - 4.2 Functionality
5. What Issues do the researchers encounter during the initial deployment of the “Click Boat: Boat Repair and Rental Management System”?
6. What are the features recommended by the respondents to improve the functionality of the system?

3. LITERATURES REVIEW

In different cases business automation is needed on a business as it grows from medium to large, Yacht repair servicing businesses and boat rentals start to use technologies to improve their performance and bring the customer to a better experience on giving the service, West Marine states that it uses its e-commerce website to test markets for new products in a cost-effective manner. [1]. the web became a famous platform over the past years; benefits that it offers are mutually equal to the side of the businesses and its customers. Web applications became famous because it does not require the user to install any additional software to access the service it is just one click away on their browser no matter what device (mobile, laptops or desktops) they are using. China yacht industry will maintain a market growth rate of no less than 30% per year, China is expected to seize the opportunity to welcome the future transfer of the yacht industry and become a new focus of the world's yacht industry[2].With the increasing numbers of the yacht, the needs for the existing yacht to repair became another likely service needed by the owners which later became the part of the YSC, with also the increasing numbers of repair to be done yacht charters start to employ repairmen to offer yacht repair and maintenance services.

On the other hand findings of IBIS World say that "The revenue of repairs and maintenance of the yacht is expected to rise over the years till the year 2021[3]". The two studies both say that the market on yacht servicing is unsaturated; owners of yacht are ready to find charters that offer yacht repair and ready to cross the borders to avail the service. Hurley states numbers and statistics regarding the needs for the yacht servicing and systems that already exist in the market that is also used by different yacht charter [4] , which also corresponds to the research during the year 2018 have conducted in china that the need and demand for e-platforms for yacht related business are high[5].Progressive web apps are now on the demand and are being popularly used nowadays because of their similarities on mobile applications, WANG, ZHENG, and CHEN findings on the research entitled Analysis of the Capital Composition for Yacht Industry in China tell that the rapid development of the yacht industry bring different problems such as difficulty to adapt the changes happening on the current market, both of Wang and his fellow researcher in China presents the existing problems such as strengthening the market with the use of technology such as cloud and the web itself to cater a wide array of services[6].

Researchers from the California surveyed 130 people who had downloaded the RideScout app to evaluate attitudes and perceptions toward mobile apps, travel behaviour, and modal shift [8]. Mobile apps and web app became a trendy way to make it easy to bring marina services on the hand of the clients and service provider, although some issues has been found such as security and privacy. To cope up with the business, service management platforms are created to deal with untidy management of transactions and data. While on the other hand adding new features to make it easy for the future end user of the system are continued to discover. Mobile apps or web apps the key here is the accessibility, the ability of the systems itself to adjust to the environment of the platform it is being access. The technology and social networking revolution in America is transforming every area of American life, including transportation. The advent of a range of new technology-enabled transportation services that take advantage of mobile communications technology has been one of the most significant shifts. Americans will have new mobility options because to technological advancements and social networking technologies. New, technology-enabled transportation services might eliminate the need for many of these services by providing Americans with more mobility options and better methods to explore these options owning a personal car is becoming more common among Americans, resulting in a considerable drop in vehicle travel.

This research wants to imply the benefits of using a Service-oriented Information system on the Boat Repair and Rental that may deal with yacht charter, local Bangka owner and tourists with the use of developmental research strategy and instrumentation such as User acceptance tests, besides, that this research also wants to gather recommendations for future improvements of the research, another goal of this Research is to develop a service oriented software that can solve problems encountered in a Boat repair and rental in terms of scheduling and management of services such as status updates of repair in email and easy report generation .Boat reservation and integrating geolocation API and features.

4. MATERIALS AND METHODS

The researchers will use the descriptive - developmental method of research in this study. Descriptive research was used to obtain information concerning the current status of the phenomena to describe "what exists" with respect to variables or conditions in a situation, as these enable statistical analysis to be made while developmental because of the proposed Service-oriented system that is to be develop. After the development phase, the Service-Oriented system will be evaluated by selected respondents. This method of research applied prominently in this study because it presented the problems the company is facing in terms of management, scheduling and report generation.

To determine the effectiveness of the software, a non-randomized type of sampling is applied to increase the accuracy of results, the chosen respondents in the research are those who have knowledge in the yacht cluster servicing and those who are in the target market of the business specified.

In developing the said service-oriented platform the researchers will use the Agile model as the framework in developing and finishing the project. This is development method is popularly used in development a system because it covers the loop holes the waterfall has such as difficulty in incremental changes.

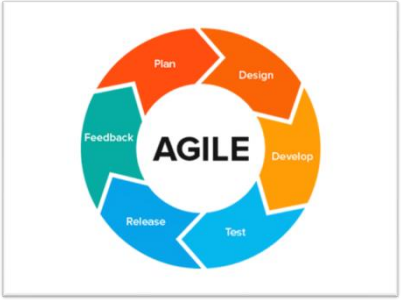

Figure 3.1: Agile Model

Figure above shows the Agile model. Agile Methodology is a people-focused, results-focused approach to software development that respects our rapidly changing world. It's centred on adaptive planning, self-organization, and short delivery times. It's flexible, fast, and aims for continuous improvements in quality [9].This method suits to the system that will be developed because this method can go back and forth to the previous and succeeding phases of the system development.

Plan

This phase is important because it will serve as the foundation of the development of the system and with this, the Researchers will do benchmarking and conduct a preliminary investigation on the existing business process of the stakeholder or client they are dealing with. In this phase also defines the scope of the system and system requirements.

Design

The process of defining the components, modules, interfaces, and data for a system to satisfy specified requirements. After the information has been gathered the Researchers will deal with the designing phase this phase includes the possible system architecture and the software tool needed to develop the said software.

Develop

This phase is more on the technical side of the process in the system, the Researchers will deal with programming and defining how to automate the business process in terms of translating it into a system requirement, in this phase also may include a prototype of the system and versions of it upon revision or reworking.

Test

After the development has been finished the output will undergo the testing process, different series of testing will be done to provide back job on the side of the developers. The Researchers will also deal on concluding what must be and what must not appear on the holistic features of the system.

Release

A release phase deals with some sort of experimental procedures even though the system passed the testing phase, it will still need to undergo on the experimentation because the environment of testing and operational is different, this phase also wants to know the feedback of the future end users of the system.

Feedback

In feedback this wants to know what to add, remove and remain on the features present on the system, the feedback also want to get the insights of the end users and deal with different issues such as data anomaly and Usability rating of the system, so that the output is ready to be deployed on the actual operational environment of the organization. after identifying the highest possible count of fish.

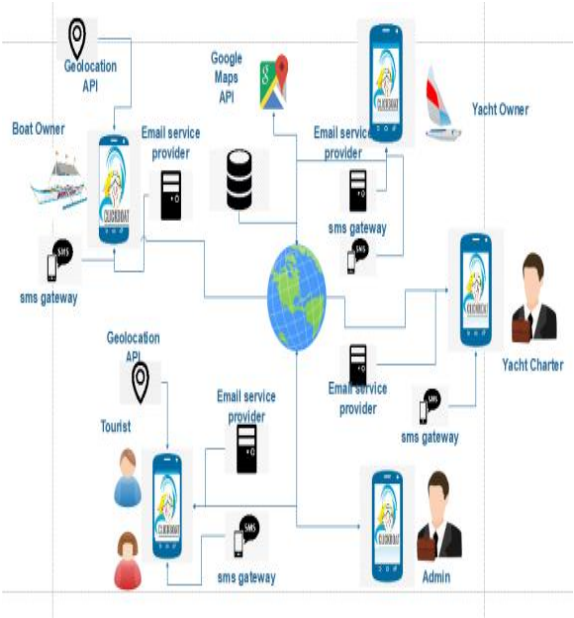

Figure 2. System architecture of the Developed Application

Table 1. Mobile App Requirements

Hardware	Technical Requirement	Purpose
Android	Android devices with Android 7+	Operating system
RAM	2gb above	Memory
ROM(Read Only Memory)	16 gb above ROM	Memory

5. RESULTS AND DISCUSSION

Problems encountered by the respondents on the existing system

Formula for getting the percentage:

Y = P% * X

Table 2. Tourist itinerary

Problem	Frequency	Rank
Finding a good boat driver for the boat ride service is difficult.	39	1st
Finding a good destination to go with is hard.	35	2nd
Finding a destination using Google search alone is difficult.	25	3rd
Searching for comments and feedbacks from other tourists is difficult.	17	4th
Others ; • price are not cascaded on boat rides	3	5th

As shown in the table above the major problem faced by the respondents in terms of tourist itinerary is "*Finding a good boat driver for the boat ride service*" which is answered by the 39 (97.5) of the respondents it can be assumed that it is an untidy job or there is no actual platform that deals with this type of business process same goes by the difficulty in "*Finding a good destination*" which is answered by 35 (87.5) of the respondents while the problem on "*Searching for comments and feedbacks from other tourists*" is answered by the 17 (42.5) respondents and using only Google maps as a platform gets 25 (62.5) answers from the respondents which can be assumed that the lack of platforms for tourist itinerary is a problem for tourists.

Table 3. Boat scheduling

Problem	Frequency	Rank
Deal with bulk transactions especially during peak seasons.	20	1st
Monitoring the good service provided by the boat owner to tourist is not recognized.	19	2nd
Managing boat reservation and scheduling transaction in manual manner.	17	3rd
Others	0	0

The respondents were asked what are the problems they have encountered in scheduling of the boat ride service for the tourist, the results as show on the table that 20 (100%) of the respondents have experienced "*difficulty in dealing with bulk transactions especially during peak season*"; while 19 (95%) of the respondents have find it difficult on "*Monitoring the good service provided by the boat owner to tourist*" and 17 (85%) of them have find it difficult in terms of "*Managing boat reservation and scheduling transaction in manual manner*".

Table 4. Yacht Repair Scheduling

Problem	Frequency	Rank
Yacht repair appointment in manual manner is difficult.	10	1st
Yacht repair appointment is challenging due to service fee are not cascaded.	10	1st
Some repair updates of yacht is not timely	8	2nd
Others	0	0

The respondents were asked to indicate what are their problems on yacht repair scheduling, 10 (100%) of the respondents answered "*difficulty in the manual process*"; 8 (80%) of the respondents have answered "*repair status update is not timely*" and also 10 (100%) have said that the "*repair scheduling is*

challenging due to the reason of service fee are not cascaded".

Table 5 . Yacht Repair management

Problems encountered	Rank
Using Log books and Record books to create a report is difficult.	1st
Difficulty to create sales invoice in a manual manner.	2nd
It takes so much time in preparing reports like annual reports, sales reports and invoice lists of finished transaction.	3rd
Difficulty in syncing information from different sources	4th

The table above shows the problems that the yacht charter have experienced using the manual system, Renato F. Lisud the manager of the Papaya Yacht Charter and Services Inc. said that although there is a system they are using in the company they find it hard to use it because it needed expertise in terms of accounting and proficiency in using flat database management systems such as Microsoft excel, one of the hindrance he also take note "the business lacks automation different clients are wanting to have the possible and easiest way to access the yacht repair services". As the researcher have demonstrate the technology solution to the Charter manager he have seen how the notification features of the system work on both Email and SMS he wants to have those kind of automation around their business saying that he wanted to implement it on the organization.

Modules of the System developed to address the common problems encountered by the respondents

Table 6. Modules of System developed to address the problem

Module name	**Access and Capability**
Tourist	• List of Destinations with current weather status, ratings and feedbacks. • List of Boat owner for boat reservation selection. • Geolocation View of Destination and current User location. • Providing rating for destinations and boat drivers • Crud functionalities on boat reservation such as Create, Cancel and update. • Receive Email and SMS notifications about transaction status. • Add proof of Down payment or Reservation fee for transaction assurance. • Password changing and retrieval. • User's profile
Boatdriver	• List of Boat reservation requests. • List of Previous Transactions and Tourist's rating. • Map view of the destination the tourist wanted. • Capability to update the Boat reservation's status. • Received SMS and Email notification about the status of boat reservation.
Boatdriver	• Verify the reservation fee or down payment of

	the tourist. • Password updating and retrieval facility. • View my profile and see data visualization of transactions
Yacht Owner	• Create, update and cancel a repair appointment. • View repair invoices. • Add repair feedback. • Receive invoice copy of repair in Email and SMS. • My profile • Password Retrieval and Updating.
Yacht charter	• List of repair appointments • Update Repair Status and generate invoices in PDF attached in the email notification to the clients. • Generate reports and view simple data visualization. • Calendar views of repairs. • View client's feedbacks. • My profile • Updating and Retrieval of Password. • Manage repairs and Services.

The table 8 shows the modules of the system the Researchers have developed to address the problems of the respondents shown in table 4, 5, 6 and 7 which describes the problem they encountered in using the manual or traditional business process. With these the Researchers have come up to the idea of creating modules that will automates the business process in terms of report generation and transaction awareness through system generated notifications that involved SMS and Email.

Tools and technology used to develop the solution to the problem

Table 7. Tools and technology used to develop the system

Tool or Technology used	**Type**	**Purpose**
Text editor tool	computer program	Edit plain text file, scripts and batch commands.
Xammp	cross-platform web server solution	Consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.
PHPmailer	Php class library file	PHPMailer is a code library to send emails safely and easily via PHP code from a web server.
DomPDF	Php class library file	HTML to PDF converter tool.
Bootstrap	CSS(Cascading Style Sheet) Framework	CSS Framework for developing responsive and mobile-first websites.
Filezilla	FTP client Software	Deploy Web app source codes such as Php,CSS and Javascript files on the Web Server.
Cpanel	Web-hosting control Panel	Manage Web apps configurations

		such as SSL certificates and other managerial task on website management,
Swing2app	Mobile App Builder	Create or Convert Web application into native mobile application.
Itextmo	SMS Gateway API	Send SMS notification in a specific mobile numbers.
OpenWeather	Weather API	Fetch weather updates and statuses.
Google Maps API	API	Fetch web mapping and geographic view of the specific location.
OpenStreetmaps	API	Create a free editable geographic database of the world.

The table above shows the technology and tools the Researchers used to develop the software in addressing the problem faced by the respondents, majority of these tools are used in creating progressive web apps and APIs are only used to give additional features on the system such as notification like Phpmailer and Itexmo does, while there are also present Geographic mapping tools used in the development such as Openstreetmaps and Google maps API.

Level of acceptance of the respondents in the developed system "Click Boat: Boat Repair and Rental Management System" in terms of?

5r=Number of Responses with 5

4r=Number of Responses with 4

3r=Number of Responses with 3

2r=Number of Responses with 2

1r=Number of Responses with 1

T=Total number of Respondents

Formula:

Mean= ((5r*5)+(4r*r)+(3r*3)+(2r*2)+(1r*1))/T

Mean2=((5r*25)+(4r*16)+(3r*9)+(2r*4)+(1r*1))/T

Standard Deviation=√ (Mean2-Mean)

Average= (∑mean)/number of parameters

Table 8. System Acceptability Rating In Terms Of User Friendliness

Parameters	Mean	Descriptive Equivalent
The system can adapt on different screen sizes.	4.68	Highly Acceptable
The UI design is learnable and easy to use/operate	4.58	Highly Acceptable
The system is easy to navigate and operate.	4.70	Highly Acceptable
The system provides different interactive views of information.	4.66	Highly Acceptable
The system provides a convenient way of password retrieval process.	4.58	Highly Acceptable
The system interface is aesthetically designed	4.66	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's user-friendliness it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's User friendliness. The traditional tourism industry is unable to follow with the development trend and with the emergence of modern personalized information services, and the service provided by the tourism e-commerce environment also important for improving the tourists' satisfaction and also the competitiveness [10].A different system has been on the trend to provide newer experience to the target market to avoid churns and decline of possible clients.

Table 9. System Acceptability Rating In Terms Of Functionality

Parameters	Mean	Descriptive Equivalent
The system provides accurate ratings, insights and data visualization	4.76	Highly Acceptable

The system improves the way the business process works.	4.63	Highly Acceptable
The system is capable of handling errors on some cases like invalid inputs such as dates and text strings.	4.63	Highly Acceptable
The system can aid the user to finish a specified task.	4.60	Highly Acceptable
The system can facilitate the accomplishment of specified tasks and functions.	4.62	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's Functionality it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's Functionality. While the incorporation of some earlier RAISA technologies such as the ATM may have radically changed the way the customer experiences [11].The concept of this information system is to automate the system even though the automation itself is not in the form of holistic automation but automating some of the most vital parts of the process such as analytics, notifications and other informatics related function that this system can bring on user and persuade to use newer systems.

Issues encountered during the deployment of the System

Table 10. Deployment Issues

Issue	Cause	Severity
DNS error	Web server Downtime	Major
MySQL error	Web server Downtime or TCP/IP socket not listening	Moderate
Web Mail Not working	email account has exceeded its quota IMAP/POP is not enabled.	Minor
Website suspension due to usage limits	Exceeded web server resources.	Minor
FTP service not working	Webserver Downtime	High
GPS not working	GPS service is not working on non-encrypted connection like HTTP (port 80)	Cosmetic
App not installed	Android version is not compatible to the app package.	Moderate
Cron Jobs Not working properly	Execution time exceeded. Incorrect permission.	Moderate
Website not safe alert	SSL certificate problem, DNS propagation in progress	Moderate

The following data on the table shows different issues that the Researcher have encountered during the deployment of the System that is according to the ISQTB (International Software Testing Qualifications Board), to sum it up, it can be seen can seen that a web based systems may encounter unexpected problems such as downtime due to multiple services and requests it is receiving across the network. In the 24/7 web industry, business owners can feel a constant need to push the latest and greatest software to the live site in order to remain competitive and ensure growth (John ,Mostashari and Mansouri,2011). It is important to always have a backup and select the best Web service package service as soon as possible because cheap services may bring drawbacks as the system scales up such as increasing volume of users and data that it process, In addition to that cross platform system such as combination of Native and mobile apps are a bit complicated due to the way they access the data and present it to the users.

System features recommended by the respondents

Table 13. Recommended features

Features	Frequency	Rank
Automated SMS and Email notification	67	1st
Payment Gateway Facility	62	2nd
Google Maps Direction	57	3rd
Simple analytics	39	4th

The implication of this finding is the respondents have suggests different features

that may improve the system for later purposes, to explain that the respondents wanted to automate almost the entire process on the business which involved notification features. The payment facility is the second majority feature they want to be added, with these result we can assume that the need of the users for automation for payment process is a must implemented feature on the system, On the Geographic features the respondents wanted to add the Google map direction services because this gives a better routes on the users while the analytics have the least number of recommendation it might be assumed that respondents have less experience or knowledge to the analytics itself.

6. CONCLUSIONS

From the above summary, the following can be concluded from this study:

- A manual process on boat repair and rental is difficult to deal especially as the data or information that the user is managing scales up especially on Boat rental where the transaction needed to process increases on peak seasons, tourists have a problem where destination to go and who else to contact to access the service. While on the other hand on yacht servicing clients that needed to avail repair services face challenges in creating a repair appointment due to price not being cascaded, paper-dependent processes, and lastly the repair management of Yacht charter is difficult because of different data sets from different flat file or paper sources that may involve logbooks, stand-alone system, and flat databases they are using, the need for automation can be seen on the problems they are facing that might resolved by the creation of a platform that can facilitate report generation and notification automation.
- The need for the development of a service-oriented platform that is intended to be used by tourists, yacht owners, boat drivers, and yacht charters where they can avail and provide services across the internet to adopt the new generation of age of paperless businesses.
- The internet itself plays a big role in the SDLC process it provides a readymade library for a fast phased development process that involves frameworks, API, libraries, and other existing systems that developers may benchmark to create innovative solutions for businesses and the community. Software severities are expected to arise on initial deployment due to incompatibility of legacy and newer systems
- Based on the findings of the acceptance test done, the study shows that it is difficult to measure the user-friendliness of the system due to the issues that users are not or does not have the same standards on Graphical interface and Cognitive understanding. The correct selection of colors, fonts and symbols must be considered to provide a user-friendly system. On the other hand, they find out that users wanted to have access to details about the information itself that may include rating and also they seek more on automation modules that may give them ease on accessing the service and the information.
- Good planning especially from a phase to phase basis such as holistic planning of deployment and procurement of resources must also discuss carefully. A series of tests from the different environments must be done first to assure the quality and assurance aspect of the system. The discrepancy between how the actual business environment and the unit testing environment is large. Software severity is a helpful tool to measure the defects and bugs of the software. In addition to that the because the deployment is one of the most vital processes of SDLC choosing the best package on technology to integrate is the best decision to do to avoid back jobs and troubles such as lack of bandwidth or performance of the system.
- The software evaluation of the study discovers that users wanted to have everything in touch of their hand; they wanted everything to be processed on just simple steps and clicks. Geographical and automated features are those features they want to have for the system improvement. There was evidence on the results that the system has a high acceptability rating on the respondents and room for its improvement is still possible especially if there is a budget for implementation and technology procurement.

7. RECOMMENDATIONS

The following are recommendations for future researches and implementation:

1.The developed system may improve its usage on a larger web hosting packaged.

2.Project budget is a very vital part of a study with these we can expand the quality of the system that we are going to develop.

3.The developers may add search box and sort the transactions based on its status and date creation.

4.Future developers may include features that can reach out other related people even though they are not the user of the system.

5.The future Researcher may use this paper as a reference related to business process automation related to Marina businesses. Accessing the system.

8. REFERENCES

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GRABage: ONLINE WASTE REPORT COLLECTION AND MANAGEMENT SYSTEM

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RATIONALE

The rapid increase in population has led to improper waste management in cities resulting in the increased spreading of diseases and pests. Waste management is necessary today because, with the rapidly growing population, waste generated by humans is also increasing. has its own manual approach to the waste management system, yet it has its limitations. Traditional waste management is using fixed schedules of when and where to collect the waste. Yet, this scheduling was rarely executed because of many reasons, such as human errors and unavoidable change of weather (rain, storm, earthquake, etc.). For those reasons, the researchers came up with making a mobile application and web-based system to notify waste collectors and prioritize barangays with urgent needs.

OBJECTIVES

The general objective of the study is to develop an online waste report collection and management system. The study attempted to identify the problems being encountered in terms of Garbage Collection Schedule and Garbage Collection Monitoring; the features needed to integrate in the proposed system; suitable hardware and software needed to develop the proposed system; proposed system that deliver a consistent result through a set of standardize procedure; level of acceptance of the proposed system based on the evaluation of the IT experts in terms of Functionality, Reliability, Portability, and Security; and level of satisfaction of the respondents of the proposed system in terms of Usability and User Experience.

METHODOLOGY

The researchers used the descriptive developmental research design, and Agile process model to make the process more flexible. In addition, the researchers used the questionnaire for measuring and comparing the level of acceptability and level of satisfaction of the end-users and IT experts on the developed online waste report collection and management system. The researchers used convenience sampling due to the on-going pandemic. The researchers used the Likert Scale to measure the holistic view and opinion to measure the level of acceptance and satisfaction of the study.

RESULTS

Based on the evaluation among respondents, the study yielded the following findings: a. The residents are having difficulties with the existing waste management in getting notifications and monitoring the collection of waste; b. several features are developed to solve the problems encountered by the residents and barangay staff; Truck Tracker and SMS notification, features that helps the waste collector and residents to easily locate the user through real-time location and be easily monitored the distance of the waste collector through SMS notification; c. the IT consultant suggested to developed Validation check in compliance with the data privacy law when asking personal information and consider the future of the system to make it reusable.

CONCLUSIONS

On findings of the study, the following conclusions were drawn; a. The researchers concluded that developed systems can be beneficial to the municipality and its residents who's having difficulties on the proper management scheduling for waste collection; b. there are so much to be done to make the system user-friendly to its customers by delivering precise results of data; c. most of the hardware and software needed are a variety of modern technologies and help the process of the development of the system in the easiest way.; d. the system passed the analysis and can deliver its purpose to the user; e. the system can perform each process based on the given task; f. the system can be utilized as an innovative and effective tool in terms of waste management schedule and garbage collection notification.

KEYWORDS *GRABage, garbage collection, scheduling system, GIS, SMS notification, information system*

1. INTRODUCTION

Every year people dump a massive 2.12 billion tons of waste. If all this waste was put on trucks they would go around the world 24 times. This stunning amount of waste is partly because 99 percent of the stuff we buy is trashed within 6 months [1].

Tons of waste is produced by people all around the world every day. The rapid increase in population has led to improper waste management in cities resulting in the increased spreading of diseases and pests. Waste management is necessary today because, with the rapidly growing population, waste generated by the human is also increasing. Hence, in the absence of an effective waste management system, it may cause health issues and natural disasters in the long run.

In the Philippines, around 35,580 tons of garbage is generated every day. Typically, each person in the country produces about half kilogram of garbage every day in the rural and urban areas, respectively. For Metro Manila, it is estimated that 8,636 tons of solid waste is generated per day, which around 0.7 kg (kilogram) per person per day due to its more modernized lifestyle. The household of our society is the biggest source of garbage in the Philippines at seventy-four (74%). Furthermore, among the total solid waste generated from households, ninety-five (95%) can still be recycled or reused (43%), or turned into compost (52%). Only 5% is made up of residuals (4%) and special/hazardous waste (1%) that are no longer biodegradable or usable [2] . In the country, the LGUs (Local Government Units) hold the primary responsibility for the efficient and effective solid waste management. Despite this law, however, poor solid waste management in the Philippines is still prevailing since controlled and open dumps are being used in the country.

Nowadays, there is existing waste collection management wherein they are using fixed schedules of when and where to collect the waste. As the dump trucks must collect every solid waste in a bin around their designated routes, the truck will be loaded. If the truck is loaded, they are obligated to unload the waste in the dumpsite and may forget to continue the collection of solid waste. So, the problem is the overflowing of neglected wastages on the roads beside the fact that they collect the solid waste rarely in a week or month. The Nasugbu, Batangas has its own manual approach to waste management system, yet it has its limitation. The waste collectors in Nasugbu are using fixed schedules of when and where to collect the waste. Yet, this scheduling was rarely executed because of many reasons, such as human errors and unavoidable change of weather (rain, storm, earthquake, etc).

For that reasons, the researchers proposed an innovation through proposing a GRABage: Online waste report collection and management system. The project is intended to create a mobile application and web-based system to notify waste collectors and prioritize barangays with urgent needs.

2. OBJECTIVES

This study aimed to develop a GRABage: Online waste report collection and management system. Specifically, the study attempts to answers the following question:

1. What are the problems being encountered in terms of:
   - 1.1 Garbage collection schedule
   - 1.2 Garbage collection monitoring
2. What are features needed to integrated in the proposed system?
3. What are suitable hardware and software needed to develop the proposed system?
4. Will the proposed system deliver a consistent result through a set of standardize procedure?
5. What is the level of acceptance of the proposed system based on the evaluation of the IT experts in terms of:
   - 5.1 Functionality;
   - 5.2 Reliability;
   - 5.3 Portability; and
   - 5.4 Security?
6. What is the level of satisfaction of the respondents of the proposed system in terms of:
   - 6.1 Usability; and
   - 6.2 User experience?

3. LITERATURE REVIEW

Based on the study of Idowu et al. (2012) entitled “Development of a Web based GIS Waste Disposal Management System for Nigeria”, developed a web-based solid waste management system for a community of Nigeria that allows residents to view the scheduled time that the garbage will be collected. It shows the system architecture, the person involved and what framework and tools was used in developed system. The system will ease the work or process of waste management of the local government units of Nigeria since they were able to monitor, manage and maintain the location of all the garbage collection tanks within the area. The study

concluded that, the use of system will achieve and ensure the cleanliness of environment and decrease the risk experiencing on improper management of waste of sources in the area of Nigeria [3].

As the population rapidly increasing, the waste produced is also growing at the same time. The country of India implemented various ways on how to solve the problem about garbage collection that is quite hard to solve since the existing system is fortuitously which leads overflowing and smelly garbage due to areas left by the collection of waste. Therefore, a study entitled "Smart Garbage Management System" by Jajoo et al. (2018) proposed a system that monitors the household garbage bin regarding the level filled. An Ultrasonic sensor is used to measure the level of garbage collected in dustbin. When the garbage level reaches a certain level, the system will send a notification to the appropriate authority via various ways. If the bins are already full or nearly full, a warning message will be generated and forwarded to the municipality via notifications, by using the WI-FI module. This will enable them in determining where and when they should go to collect waste. This keeps track of the effort of checking out the area by going there. The system aid both citizens and the Brihan Mumbai Municipal Corporation (BMC) which usually fails to gather the waste of some areas. When a notice is ignored by a lower authority, the following notification is sent to the higher authority. This allows us to efficiently handle waste. Additionally, an Android application is being developed to assist garbage collectors in informing the authorities when waste is being collected. End- to-end communication is established as a result of this [4].

A study entitled "Automated Garbage Collection Alert System" developed a mobile application to manage garbage in various locations based on the analysis of a technique for the optimization of the waste collection and transportation system, which was based on Geographic Information System (GIS). Through this mobile application, it allows communicating directly with the Municipal Council via this smartphone application and receive information about the garbage dump's specific location. It focuses on waste dumps in the streets and main roads in this system. People and local councils in a nation like Sri Lanka can save time and money by utilizing thissmartphone application [5].

4. MATERIALS AND METHODS

The researchers used the descriptive-developmental method in the study. Descriptive research uses a quantitative research method by collecting quantifiable information concerning about the current status of the study. The purpose of using descriptive method is to determine average and frequencies of the data and information gathered. While developmental method is to be used since this study introduced a waste management system named GRABage: Online Waste Report Collection and Management System

The system used the Agile process model in developing and finishing the waste report and management system. The model consists of five cyclical phases: Brainstorm, Design, Development, Quality Assurance and Deployment.

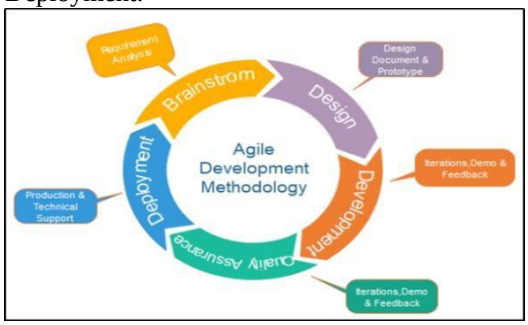

Figure 1. Agile Process Model

Figure 1 shows the Agile Software Development Life Cycle (SDLC). It is an SDLC that focuses on adapting changes that became popular because of its effectiveness and result-oriented approach. This model is suited for the developed system since it focuses on the iterative development.

Brainstorm Phase

The brainstorm phase often referred to as the requirements gathering phase wherein the researchers determined the functions and requirements needed in the study. The researchers also gathered information in order to analyze the business requirements and system deliverables.

Design Phase

In the design phase, visual design and the architectural structure of the app are the two ways to approach this phase. The researchers determined the features of the system and describe operation and flow where it has a goal to be user-friendly.

Development Phase

The development phase is also known as construction stage where writing codes and converting design into actual software occur. The researchers developed the system using programming process, codes and commands, in order to meet business requirements and desire deliverables.

Quality Assurance (QA) Phase

In the quality assurance phase or the testing phase, the researchers will examine the project's performance and making sure that it is bug-free software. Series of test will be conducted and researchers will analyze project to ensure the code is clean and business goals are met.

Deployment Phase

In the deployment phase, the researchers issue the project to the user's work environment – either for demo or actual use. Further iteration happens when there is a new introduces features on the software. The system was evaluated and its effectiveness and quality was assessed.

Figure 2. Waste Calendar on Admin Module

Figure 2 displays the waste calendar on the Admin module. On this interface, admin can manage waste schedule. There are five (5) different status of waste collection and it represents different colors. Orange color is for Pending status, blue is for On schedule, the collected status as green, Gray for Unconfirmed status and red as Denied. Schedules can be added in two different ways, importing files to add more than one schedule or setting individual schedule for a specific barangay. On the left side of the screen, navigational menu is added and the menu includes Waste Calendar, Waste Schedule List, Notification, Users, Settings and Logout button.

Figure 3. Verification Code on Resident Module

The figure 3 shows the message including the verification code that was sent when resident sign up to GRABage. To successfully register, resident should enter code correctly to the application.

Figure 3. Verification Code on Resident Module

As shown in the figure 4, residents will be able to capture image to send request for prioritization of their barangay and the image will serve as proof for faster verification of the request.

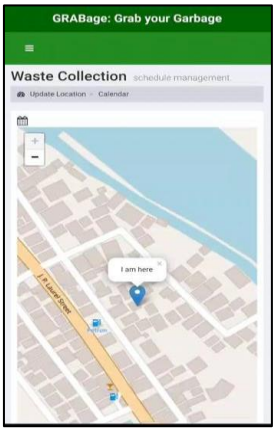

Figure 5. GPS Location on Resident and Staff Module

Figure 5 displays the real-time location of the residents and staff when "Update Location" button is clicked. It shows the specific location of the user using map. Beside that button, "Calendar" button is located to view detailed waste schedule of their barangay.

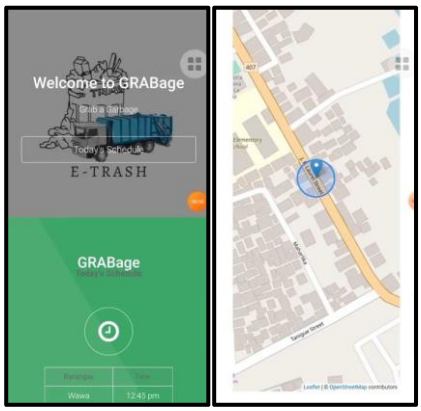

Figure 6. Truck Tracker

On the figure 6, it shows the module for truck tracker. The left image shows the truck waste collector that can view barangays scheduled for today's collection. When the truck started collecting waste, real-time location of the truck is display on the application, as shown on the right image.

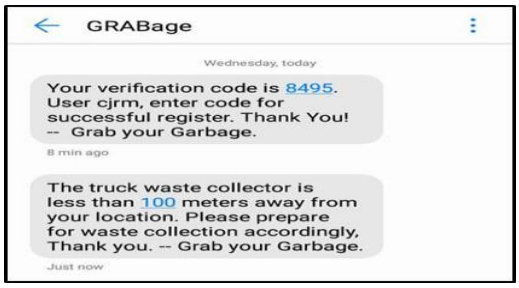

Figure 7. SMS Notification

For the figure 7, it displays the message that was sent to users with today's schedule and the truck waste collector is less than 100 meters away from their location. The purpose of this message was to notify users about the distance of the truck and they can have time to collect and gather their waste and to pile outside while waiting to be collected.

Hardware and Software Requirements

Reflected on the Table 1 and 2 are the hardware and software needed in developing the GRABage: Online Waste Report Collection and Management System.

Table 1. Hardware Requirements

Particulars	Specification
Processor	2.30 GHz
RAM	4 GB RAM
Storage	500 GB
HDD	500 GB or more
Input Devices	Keyboard/Mouse
Output Devices	Monitor

Table 2. Software Requirements

Particulars	Software
Operating System	64-bit Operating System
Database	MySQL
Programming Language	PHP
Web Editor	XAMPP
Web Technology	HTML, CSS

5. RESULTS AND DISCUSSION

In the preparation and evaluation stage, the researchers conducted the evaluation procedure of the developed system to the fifty (50) respondents comprising of thirty (30) residents of Nasugbu, fifteen (15) Barangay Staff and five (5) IT Experts. The survey questionnaires made by the researchers were validated and used to answer the following question.

The researchers provided two (2) sets of survey questionnaires, one for the end users (barangay staff and residents) that seek to answer the evaluation forms to determine problems being encountered in terms of garbage collection schedule and garbage collection monitoring and also to determine level of satisfaction in terms of Usability and User Experience which is conducted personally and another set of questionnaires through Google forms for the IT experts that seek to answer the evaluation forms to determine level of acceptance of the developed system in terms of Functionality, Reliability, Portability and Security. Also, they were asked to answer the evaluation forms to determine level of satisfaction of the end users of the developed system in terms of Usability and User Experience.

Table 1. Problems Encountered by the End users Relative to the Existing Garbage Collection Schedule

Problems	Frequency	Rank
Schedules are not being followed because of waste truck collectors are limited.	36	2
There is no rescheduling method after the waste collectors failed to collect waste from certain location.	41	1
Users are not updated when there are changes in the waste collection schedule.	30	3

Table 1 presents the problem encountered by the end users relative to the existing garbage schedule. Based on the result stated on the table above, "There is no rescheduling method after the waste collectors failed to collect waste from certain location" got the highest rank. The researchers find that the users aren't able to request another schedule

when the waste collectors failed to collect. The user must be able to request rescheduling to avoid their waste piling up and being neglected. Related to the study of Idowu et al. (2012) entitled "Development of a Web based GIS Waste Disposal Management System for Nigeria" that focuses in providing residents the scheduled time that the garbage will be collected to decrease the risk of improper management of waste.

Table 2. Problems Encountered by the End users Relative to the Existing Garbage Collection Monitoring

Problems	Frequency	Rank
Collected waste is not monitored by MENRO.	36	2
Unmonitored overflowing of garbage in barangay results unpleasant pile of waste in roads and bins.	37	2
Detection and management of waste is one of the primary problems of the municipality.	42	1

Table 2 shows the problems encountered by the end users relative to the existing garbage monitoring. Based on the table above, "Detection and management of waste is one of the primary problems of the municipality" got the highest rank. This clearly states that one of the primary problems of the municipality is the detection and management of waste. Most of barangay relay on municipal waste collector. So, if the waste of different barangays isn't collected it might cause some issues such as health issues or piling of waste on roads and bins. That is also the reason why the India created the "Smart Garbage Management System", studied by Jajoo et al. (2018), that propose a system that monitors the household garbage bin regarding the level of waste filled.

Table 3. Features Needed to Integrate in the Developed System

Feature	Function	Description
Calendar	The Calendar is added to help the admin to easily see the schedule for the month, week, or day. With this, the admin can be easily aware about the status collection at a certain time.	A series of pages or charts of days, weeks, months, or specific years. This may contain schedules and holidays.
Verification Code	Through the help of SMS notification, this feature sends randomly generated numbers to the user's personal number. This helps the application to be more secure and reliable when signing up to the system.	The Verification code is an OTP (One Time Pin) that is randomly generated to let users have security protection.
Image Capturing	The Camera Capture feature is designed to capture the report of end-users, reports like piling trash or uncollected waste in their barangay. This report will be used to let the residents request for collection on their place. This will allow the admin to identify the validity of the requests of the users.	Image capturing is an act of recording and/or capturing an image or other information using a computer or camera. This is a technology designed for capturing visual images.
GPS location	The GPS location feature is added on residents' modules. This feature updates the location of the user through a button, and then will also be used for waste collection routes.	A GPS Location is widely used to navigate someone's exact location anywhere around Earth's surface.
Truck Tracker	The truck tracker is a real-time location feature that helps the end users to see the location of the truck and with the integration of SMS notifications, residents get notified when truck waste collector is 100 meters away from their location.	Truck Tracker is used to keep track of the real-time location of a specific truck.
SMS Notification	Not all users have the most advanced smartphones. The SMS Notification feature of our developed system made it possible to let end users is updated about the truck collector's distance away from on-scheduled barangay.	A SMS Notification is a short message sent in response to an event, can be an update or service alert.

Table 3 shows the features needed to integrate to the developed system and their corresponding descriptions on how it is used in the system. Moreover, these features help the researchers to achieve the functionality and purpose that the system should deliver to its customers. The most highlighted features in the system are the truck tracker and SMS notification and it is also highlighted on the study of Ghazali et al. (2019) about "Mobile Application on Garbage Collector Tracker Using Google Maps" that the project was designed to alert communityon garbage duty schedule, be notified and be able to track the location of garbage collector using Google maps application [6].

The following are the hardware needed to run the web application.

Table 4. Hardware Specification for Web Application

Items	Requirement	Description
Processor	2.30 GHz	A better processor can process basic command and can help in better performance of the system.
Memory/ RAM	4.00 GB	For the better speed and performance of our application, the researchers require at least 4GB RAM
Storage	500 GB	The researchers require at least 500GB of storage capacity to ensure the best performance of the application
Input device	Keyboard, Mouse	The input devices help the user to input their basic and required information. The input devices can help in better processing of data.
Output device	Monitor	The output device/s can help in producing better way to let end users interact through the application
Communication Interface	Mobile device, Modem	The communication interface helps the end users to have better communication.

Table 4 shows the hardware needed to run the web application. It also shows the requirement specification and its description of use of each hardware. To acquire better performance of the web application, a better processor, higher storage and at least 4 GB memory are recommended by the researchers. To avoid hassle using the system, researchers suggested to use keyboard and mouse as input devices and a monitor as an output device. And also for easiest access to the web application, mobile device and modem are suggested as communication interface.

The following are the hardware needed to run the mobile application.

Table 5. Hardware Specification for Mobile Application

Items	Requirement	Description
Processor	1.0 GHz	A better processor can process basic command and can help in better performance of the system.
Memory/ RAM	2.0 GB	For the better speed and performance of our application, the researchers require at least 4GB RAM
Storage	16 GB	The researchers require at least 500GB of storage capacity to ensure the best performance of the application
Display	5.5 inches, 720 x 1280 pixels	A better display can help to produce better way to let end users interact through the application
Communication Interface	Mobile device, Modem	The communication interface helps the end users to have better communication.

Table 5 shows the hardware needed to run the mobile application. It also shows the requirement specification and its description of use. To acquire better performance of the mobile application, a better processor, higher storage and at least 2 GB memory are recommended by the researchers. To avoid hassle using the system, researchers suggested to use a 5.5 inches, 720 x 1280 pixels for better display and interaction between end user and mobile application. And also for easiest access to the mobile application, mobile data and modem as communication interface.

The following are the software needed to run the web application.

Table 6. Software Specification for Web Application

Items	Requirement	Description
Web Browser	Google Chrome, Microsoft Edge	Using the required web browser can have a better percentage of the performance success
Operating System	64bit OS, Android version	The required operating system can help the end users to enjoy the technology feature of the application such as Real-Time Location and SMS Notification.

Table 6 shows the recommended software for the implementation of the developed system. To acquire smooth interaction of web application and end user using their personal devices, 64bit OS and android version is recommended to enjoy the major features of the application. And for better performance, the required web browser such ass Google Chrome and Microsoft Edge is recommended.

Table 7. Software Specification for Mobile Application

Items	Requirement	Description
Operating System	Android Lollipop above	The required operating system or above can help the end users to enjoy the technology feature of the application such as Real-Time Location and SMS Notification.

Table 7 shows the recommended software for the implementation needed to run the mobile application. The mobile should be Android Lollipop above to smoothly run the system on mobile devices and can definitely enjoy the major features of the application.

Table 8. Results from Quality Assurance Analysis

Module	Scenario	Comments	Result
Admin Module	Login	Security services must be develop to prevent unauthorized access to, modification of private date (data).	Passed
	Adding of inputs like dates, time and reason for declines	The system should be able to handle negative inputs and testing of data.	Passed
	Notification	Notifications should notify the admin in all status even offline	Failed
	Prioritization Request	Set cut-off time to provide reminders on the inquiry logged after the cut-off time that this will be entertained on the next working day to avoid delays and backlogs	Failed
Staff and Resident Module	End users inputs private information	Provide protection for all gathered private information of all end users.	Passed
	Reminder and Tips	Provide guidelines, rules and regulation that both parties will use the software properly. Give assurance between waste management and residents about the maintenance, updates and future of the system.	Failed

Table 8 shows the components tested by the Quality Assurance (QA) analyst which are the admin module, staff and resident module and the truck tracker application. Using a Test Case Document and video presentation, QA analyst discovers bugs and errors that might not satisfy users' requirements. Scenarios like login, adding inputs like dates passed the evaluation on the admin module and entering private information for the staff and residents. However, failed results encountered in notification and prioritization request on the Admin Module. Notification should be able to notify admin even offline and setting of cut-off time to be entertained from the residents. On the Staff and Resident Module, failed results include the reminder and tips. This should include guidelines and regulations on how to use the system.

Table 9. Level of Acceptance of the Developed System based on the Evaluation of the IT Experts

Parameters	Weighted Mean	Verbal Interpretation
Functionality	4.60	Perfectly Acceptable
Reliability	4.65	Perfectly Acceptable
Portability	4.80	Perfectly Acceptable
Security	4.70	Perfectly Acceptable
Combined Weighted Mean:	**4.71**	**Perfectly Acceptable**

Table 9 shows level of acceptance of the developed system based on the evaluation of IT Experts. Based on the result of the evaluation, portability got the highest weighted mean and gives conclusion to the researchers about how does the system is portable and can adapt to different kinds of environment. It is supported with the study of "Web-Based Toxic Waste Management Information System Using Unified Approach Method" that accelerates the method of manual recording and reporting of waste management and includes the benefits of the system in overcoming the existing management of waste [7].

Table 10 Level of satisfaction of the end users to the developed system

Parameters	Weighted Mean	Verbal Interpretation
Usability	4.60	Extremely Satisfied
User Experience	4.79	Extremely Satisfied
Combined Weighted Mean:	**4.70**	**Extremely Satisfied**

Table 10 shows the level of satisfaction of the end users to the developed system in terms of Usability and User Experience. Based on the result, end users are extremely satisfied to the developed system. This means that the users easily get familiar with the developed system and to learn how to use it properly. It is

supported with the study of Idowu et al. (2012) entitled "Development of a Web based GIS Waste Disposal Management System for Nigeria", that aims to ease the experience of the end users while performing great usability.

Table 11. Level of satisfaction of the IT Experts to the developed system

Parameters	Weighted Mean	Verbal Interpretation
Usability	5.00	Extremely Satisfied
User Experience	4.76	Extremely Satisfied
Combined Weighted Mean:	**4.88**	**Extremely Satisfied**

Table 11 shows the level of satisfaction of the IT Experts to the developed system in terms of Usability and User Experience. Based on the result, IT Experts are extremely satisfied to the developed system.

6. CONCLUSIONS

This part shows all the conclusions that the researchers gathered during the development and evaluation of GRABage.

1. Based on the findings, the researchers concluded that GRABage application can be beneficial to the municipality and its residents having difficulties on the proper management scheduling for waste collection.
2. Given the different features implemented to the developed system, researchers have concluded that (1) Image Capturing can be beneficial to prove the request of the residents about the prioritization, (2) Calendar that allows admin to create manageable schedule of waste collection, (3) Verification code that ensure securities for those who are signing up in the system, (4) GPS Location that the residents can use to save their location (5) Truck tracker which is to track the distance of the truck to the residents and it is notified through (6) SMS notification.
3. The researchers have concluded that most of the hardware and software needed to integrate to the developed system are a variety of modern technologies and helps the process of the development of the system in easiest way.
4. The researchers concluded that in terms of main features implemented, the system passed the analysis. For this, it is concluded that the system can deliver its purpose to the user which is to manage schedule for waste collection. However, the system fails on providing the assurance to the future of the system in terms of handling notifications, prioritization request and the reminder and tips interface. With this, improvements should be added like how notification should be able to notify admin even offline and setting of cut-off time to be entertained from the residents' sent prioritization request and for the reminder and tips, this should include guidelines and regulations on how to use the system.
5. Given the high ratings and evaluation of acceptance of the IT Experts to the developed system, the researchers have concluded that the quality of the developed system conforms to the ISO/IEC 25010 standards. Researchers also concluded that the system can perform each processes based on the given task.
6. With the high ratings of both the end users and the IT experts on the level of satisfaction to the developed system, the researchers concluded that the system can be utilized by the local unit of Nasugbu as an innovative and effective tool in terms of waste management schedule and garbage collection notification.

5. RECOMMENDATIONS

The following are the recommendations for the enhancement of the GRABage: Online Waste Report Collection and Management System.

1. The developed system is recommended by the researchers to be useful to the municipality to improve waste management collection in different barangays preventing one of the main problem which is managing and disposal of waste to ensure that garbage scheduling and monitoring are achieve in faster and easier way.
2. The integration of Heatmap to view barangays that have the most waste per collection and improvements for dashboard to have weekly or monthly generation of data displayed are highly suggested to be included on the developed system.
3. The developed system should provide step by step procedure or tutorial of how to use the system especially for new users.
4. The inclusion of proper presentation of data in printing of reports is also recommended to identify the formality of the documents.

5. For better handling of reports in tables, the developed system should be able to sortstatus on their choice.
6. The GRABage application should have an IOS and Android application that can be easily downloaded from the App Store and Play Store.
7. The research team recommended on giving assurance that the developed system should be reusable in the future and maintain connection between the waste managementand resident users.
8. For future researchers, the research team recommend to enhance the system, make it appealing to the users and more user-friendly.

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